Ref. [59] — AdaGrad Avoids Saddle Points

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1. The research problem

Classical GD has one fixed scalar step size, whereas AdaGrad changes the geometry of its updates using the *history of observed gradients*. Can a stable-manifold argument still show that adaptive methods avoid saddle points almost everywhere? The paper addresses this for the AdaGrad family, including scalar, diagonal and full-matrix preconditioning.

2. First principles: AdaGrad is a changing metric

For a simple diagonal AdaGrad rule, define the accumulated squared-gradient vector s_k and preconditioner D_k:

$$s_k = s_{k-1} + g_k \odot g_k, \qquad D_k = \operatorname{diag}\big((s_k + \delta \mathbf{1})^{-1/2}\big), \qquad x_{k+1} = x_k - \eta D_k g_k.$$

These familiar equations are a *teaching representation* of diagonal AdaGrad; the paper analyzes more general variants. The preconditioner gives different effective steps to different coordinates, which can reduce the conditioning problem but complicates dynamical systems analysis because D_k depends on the entire trajectory. At a strict saddle x^* with Hessian H, a fixed positive-definite matrix D would induce a local linearization

$$e_{k+1} \approx (I - \eta DH) e_k.$$

Even if DH is not symmetric, it is similar to $D^{1/2} H D^{1/2}$, which is symmetric and has the same inertia as H (for $D \succ 0$). Thus negative curvature still yields an unstable mode. The challenge is that *actual* AdaGrad has a time-dependent, history-dependent D_k.

3. Method and main results

The authors develop step-size stabilization arguments and a recursive representation for the adaptive preconditioner, allowing center-stable manifold methods to be applied despite the gradient-history dependence. They establish almost-everywhere saddle avoidance for AdaGrad variants, including scalar, diagonal and full-matrix versions, under the required regularity and initialization conditions.

Principal result. AdaGrad's changing coordinate scales do not inherently destroy the strict-saddle avoidance phenomenon: the trajectories analyzed in the paper avoid saddles from almost all initializations.

4. Why this is not just an application of the GD theorem

In GD with fixed η, the update $G(x) = x - \eta \nabla f(x)$ depends only on the present state. For AdaGrad, the state is better represented as *parameters plus optimizer history*. Two runs at the same x but with distinct accumulated gradients may take different next steps. A theorem for constant D therefore does not automatically apply: the authors' novel work is in controlling this additional dynamical state.

5. Assumptions and survey relevance

The theorem uses the paper's differentiability, step-size stabilization and almost-everywhere initialization hypotheses. It is an asymptotic avoidance statement, not a worst-case iteration bound for escaping a saddle; it also does not automatically transfer to Adam, RMSProp, large-step training or nonsmooth neural-network objectives. Global optimality requires a separate landscape theorem.

Scope. "AdaGrad avoids saddles" under the specified theory does not mean an adaptive preconditioner guarantees efficient escape from every degenerate saddle.

This is particularly relevant to **RQ2** (adaptive dynamics), **RQ3** (negative curvature versus avoidance), and **RQ4** (modern-optimizer theory).

6. Our notes

This paper is a useful bridge from the GD saddle-avoidance results to adaptive optimizers. AdaGrad is harder to analyze because the update remembers past gradients, but that history does not automatically

make saddle convergence likely. The result is reassuring for AdaGrad specifically, without claiming that the same theory already covers every Adam-style optimizer used in deep learning. **Original source:** https://proceedings.mlr.press/v162/antonakopoulos22a.html. **Verification:** official ICML/PMLR version; diagonal update and fixed-metric example are pedagogical.